

Computability & Logic, Exercises to Lectures 5+6

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Theoretical Exercises

*We switch to the book by Huth and Ryan, *Logic in Computer Science*!*

While all exercises are recommended, we suggest completing 1.2.1, 1.2.3, 1.4.2, 1.4.5, 1.5.1, and 1.5.2 before the rest.

Proofs in Natural Deduction

Of course, you can do the proofs in natural deduction by hand, but it is maybe more fun to use a tool for it. One of our intrepid TAs has developed a very nice tool for this process: <https://github.com/logicontainer/logicbox/> where you can type a Fitch-like proof in natural deduction. The webservice will check your proof for correctness! A tutorial is available on the github page.

We have already set up the exercises below, so you could directly jump in. In particular, click on the below hyperlinks to reach the preloaded exercises.

- 1.2.1 (c, k, o, q)
- 1.2.3 (a, d, t, v)
- 1.2.5 (a)

Truth Tables

- 1.4.2 (a,d,i). Also: indicate if the formula is valid and/or satisfiable.
- 1.4.5
- 1.4.12 (a,c,e)

Semantics

- 1.5.1
- 1.5.2
- 1.5.6 (g)